The Daily Guide Road Map

Role: AI Engineer Intern (Azure Stack)

Duration: 12 Weeks (3 Months)

📅 Daily Routine Structure

To succeed, the intern must follow this daily structure:

- **1 Hour Theory:** Read the official Microsoft Docs linked in the daily plan.
- **2 Hours Practice:** Write code, fail, debug, and fix.
- **5 Hours Project:** Build the weekly deliverable.
- **End of Day:** Push code to GitHub and document what you learned.

🔵 MONTH 1: THE FOUNDATION (Python & Azure ML)

Theme: "Mastering the Tools of the Trade"

✅ WEEK 1: Python Data Science Bootcamp

Goal: Go from an empty laptop to predicting housing prices.

Monday: The Professional Setup

Theory: Understand Virtual Environments (Isolation), Git (Version Control), and Cloud Basics.

Practice:

Install VS Code + Python Extension + Azure ML Extension.

Install Miniconda. Create env: conda create -n azure-ai python=3.10.

Create a GitHub repo azure-ai-intern-journey and clone it locally.

Create a free Azure Account.

- **References:** Git Handbook | Conda Guide | Azure Free Account

Tuesday: Data Wrangling with Pandas

Theory: DataFrames vs Series, Data Cleaning strategies.

Practice:

Load the "House Prices" CSV using pd.read_csv().

Check for nulls: df.isnull().sum().

Fill missing values: df.fillna() (mean for numbers, mode for text).

Filter data: Select houses with > 3 bedrooms.

- **References:** Pandas Getting Started
- **Exercise:** Pandas 100 Exercises
- **Video:** Pandas Tutorial (Corey Schafer)

Wednesday: Data Storytelling (Visualization)

Theory: Distributions, Correlations, Outliers.

Practice:

Use Seaborn (sns.heatmap) to find which features correlate with Price.

Use Matplotlib (plt.hist) to see the distribution of prices.

Identify outliers using Boxplots.

- **References:** Seaborn Gallery
- **Video:** Data Visualization with Seaborn

Thursday: Your First Model (Scikit-Learn)

Theory: Supervised Learning, Features (X) vs Target (y), Train/Test Split.

Practice:

Split data: train_test_split(X, y, test_size=0.2).

Import LinearRegression from sklearn.linear_model.

Train: model.fit(X_train, y_train).

Predict: preds = model.predict(X_test).

- **References:** Scikit-Learn Linear Regression
- **Video:** Machine Learning in 5 Minutes

Friday: Evaluation & Metrics Theory: Why accuracy is bad for regression. MAE vs RMSE.

Practice: Calculate Mean Squared Error (MSE) and R2 Score.

Save your model using joblib.dump().

- **References:** Regression Metrics Guide
- **Video:** MSE, MAE, RMSE Explained

🏆 **Deliverable:** A committed Jupyter Notebook that cleans data and predicts prices.

✅ WEEK 2: Advanced ML & Feature Engineering

Goal: Handling real-world messy data and classification.

Monday: Feature Engineering

Theory: Machines only understand numbers. Categorical data issues.

Practice: Use OneHotEncoder for text columns (e.g., "Color: Red" -> [1,0,0]).

Use StandardScaler to scale numerical data (Salary vs Age).

Build a preprocessing function.

- **References:** Preprocessing with Sklearn

Tuesday: Classification (The "Yes/No" Problem)

Theory: Logistic Regression (Sigmoid function), KNN distance.

Practice:

Switch dataset to "Customer Churn" (Target: 1=Left, 0=Stayed).

Train LogisticRegression.

Generate a Confusion Matrix to see False Positives.

- **References:** Classification Metrics
- **Video:** Logistic Regression Intuition

Wednesday: Tree-Based Models

Theory: Decision Trees, Random Forest (Ensembling), Overfitting.

Practice: Train a RandomForestClassifier.

Compare training accuracy vs testing accuracy to check for overfitting.

Visualize one tree from the forest.

- **References:** Ensemble Methods Guide

Thursday: Model Comparison Framework

Theory: No Free Lunch Theorem.

Practice: Write a Python loop to train 3 models: LogReg, RandomForest, SVM.

Store their accuracy scores in a dictionary.

Print a leaderboard: "Best Model is X with Y% accuracy".

Friday: Hyperparameter Tuning Theory: Grid Search vs Random Search.

Practice: Use GridSearchCV to find the best n_estimators for Random Forest.

Refine your model with the best parameters.

- **References:** Hyperparameter Tuning Sklearn
- **Video:** Grid Search vs Random Search

🏆 **Deliverable:** A "Model Selection Script" that automatically picks the best algorithm.

✅ **WEEK 3: Transition to Cloud (Azure ML Studio)**

Goal: Moving code from Laptop to Azure Cloud.

Monday: Azure Core Resources

Theory: Resource Groups, Regions, Storage Accounts.

Practice: Create a Resource Group.

Create an Azure ML Workspace.

Explore the "Studio" UI (https://www.google.com/search?q=https://ml.azure.com).

- **References:** Azure Resource Groups | Azure ML Quickstart

Tuesday: Cloud Compute

Theory: Compute Instances (VMs) vs Compute Clusters (Scalability).

Practice: Create a standardized "Compute Instance" (Standard_DS11_v2).

Create a "Compute Cluster" with min_nodes=0 (to save money).

- **References:** Azure ML Compute Concepts

Wednesday: Data Assets

Theory: Data Versioning. Why don't we just upload CSVs manually?

Practice: Upload the Churn CSV to Azure Blob Storage.

Register it as a Data Asset in Azure ML (v2).

Consume the data in a notebook using ml_client.data.get().

- **References:** Azure ML Data Assets

Thursday: Cloud Notebooks

Theory: Managed Environments.

Practice: Clone your GitHub repo into the Azure ML Studio file system.

Run your Week 2 code on the Cloud Compute Instance.

Friday: Jobs (The SDK v2)

Theory: Command Jobs. Detaching execution from the IDE.

Practice: Convert your notebook to a clean train.py script.

Use the Python SDK v2 (command()) to submit the script as a Job to the Cluster.

- **References:** Azure ML Jobs SDK v2

🏆 **Deliverable:** A green "Completed" Job in the Azure Jobs tab.

✅ **WEEK 4: Automation & AutoML Goal:** Letting Azure do the heavy lifting.

Monday: AutoML (UI)

Theory: Automated Machine Learning concepts.

Practice: Use the "Automated ML" tab in Studio.

Point it to your Churn Data Asset.

Run it and watch it test 50+ models.

- **References:** AutoML Studio Tutorial

Tuesday: AutoML (SDK)

Theory: Infrastructure as Code.

Practice: Configure an automl.classification job using Python.

Set limits (timeout, primary metric).

Retrieve the best model programmatically.

- **References:** AutoML SDK v2 Guide

Wednesday: Pipelines Theory

Theory: MLOps, DAGs, Reproducibility.

Practice: Draw a pipeline diagram: Data Prep -> Train -> Evaluate.

Define inputs and outputs for each step.

- **References:** ML Pipelines Concept

Thursday: Building Pipelines

Theory: The @pipeline decorator (DSL).

Practice: Create two components: prep_data.py and train_model.py.

Stitch them together in a pipeline job.

Run the pipeline.

- **References:** Create Pipelines in Python

Friday: Model Registration Theory: Artifact Management.

Practice: Take the output of your pipeline (the .pkl file).

Register it in the Model Registry with version 1.

- **References:** Model Management

🏆 **Deliverable:** A repeatable Pipeline that ends with a Registered Model.

🟠 **MONTH 2: DEPLOYMENT & AI SERVICES Theme:** "Production Engineering"

✅ **WEEK 5: MLOps & Responsible AI**

Goal: Governance and Fairness.

Monday: Model Registry Deep Dive

Practice: Tag models (stage="staging", accuracy="0.85"). Load a specific version back into a notebook.

Tuesday: RAI Dashboard Practice: Add the "Responsible AI" component to your pipeline. View the generated dashboard.

- **References:** Responsible AI Dashboard Docs

Wednesday: Explainability Practice: Use "Feature Importance" to see why users churn (e.g., is it Price or Support calls?).

Thursday: Environments Practice: Create a custom Environment (Docker context) with a specific conda.yml file.

- **References:** [Manage Environments in Azure ML]

Friday: Fairness Audit

Practice: Check if the model performs worse for specific demographics (simulated).

✅ WEEK 6: Real-Time Deployment (APIs)

Goal: Integrating models into apps.

Monday: FastAPI Basics Practice: Build a local web server (app.py) with one route /predict that takes JSON and returns a result.

- **References:** [FastAPI Official Tutorial]

Tuesday: Managed Online Endpoints

Practice: Create an Endpoint in Azure ML (the "server").

- **References:** [Deploy Online Endpoints]

Wednesday: The Scoring Script

Practice: Write score.py. It needs two functions: init() (load model) and run(data) (predict).

Thursday: Deployment

Practice: Push the deployment configuration (Model + Environment + Score.py + Compute). Wait for blue/green deployment.

Friday: Testing & Auth

Practice: Use curl or Postman to send data to the HTTPS URL. Use the Key for authentication.

🏆 **Deliverable:** A live API URL that you can ping from your laptop.

✅ WEEK 7: Azure AI Services (Vision & Speech)

Goal: Cognitive Services (No training required).

Monday: Computer Vision

Practice: Create a Vision resource. Use the "Image Analysis 4.0" API to detect objects and generate captions for images.

- **References:** [Azure AI Vision Quickstart]

Tuesday: OCR (Read API)

Practice: Send a photo of a receipt. Extract the text lines and coordinates.

Wednesday: Speech Services

Practice: Send text to the TTS (Text-to-Speech) API to generate an .mp3 file.

- **References:** Azure TTS Quickstart

Thursday: Integration Script

Practice: Write a script: Image -> OCR -> Text -> TTS (Audio). "Reading a sign out loud."

Friday: Mini-Project

Practice: Build a "Receipt Scanner" that outputs a JSON summary of expenses.

✅ **WEEK 8: Document Intelligence & Search**

Goal: Knowledge Mining.

Monday: Document Intelligence

Practice: Use the "Layout" model to extract tables from a PDF invoice.

- **References:** Document Intelligence Overview

Tuesday: Custom Extraction

Practice: Use the Labeling Tool to train a custom model on a specific form type.

Wednesday: Azure AI Search Setup

Practice: Create a Search Service. Define an Index schema (ID, Content, Title).

- **References:** Azure AI Search Tutorial

Thursday: Indexing Data

Practice: Connect a Blob container (with PDFs) to the Search Indexer. Run the indexer.

Friday: Querying Practice: Send search queries to the index using the Python SDK. Implement "Fuzzy Search".

🏆 **Deliverable:** A Python script that searches through 50 PDFs and finds relevant paragraphs.

🟣 MONTH 3: GENAI & ARCHITECTURE

Theme: "The Modern AI Stack"

✅ WEEK 9: Azure OpenAI & Prompt Engineering

Goal: GPT-4 Mastery.

Monday: Azure OpenAI Studio

Practice: Apply for access. Create a resource. Deploy gpt-35-turbo and gpt-4o.

- **References:** Azure OpenAI Concepts

Tuesday: Prompt Engineering

Practice: Test "Zero-shot" vs "Few-shot" in the playground. Save the best prompts.

- **References:** Prompt Engineering Guide

Wednesday: System Messages

Practice: Create a "Rude Pirate Bot" using only the System Message to control tone.

Thursday: Python Integration

Practice: Use OpenAI.AzureOpenAI. Write a loop to have a conversation in the terminal.

- **References:** Azure OpenAI Python SDK

Friday: Chatbot History

Practice: Manage the "Context Window". Append user/assistant messages to a list before sending to the API.

✅ WEEK 10: RAG (Retrieval Augmented Generation)

Goal: Chat with your Data.

Monday: Embeddings

Practice: Deploy text-embedding-3-small. Convert text strings into vector arrays.

- **References:** Embeddings Guide

Tuesday: Vector Search

Practice: Enable Vector Search in Azure AI Search. Create a Vector Index.

- **References:** Vector Search Concepts

Wednesday: The RAG Flow

Practice: Coding the logic: User Question -> Embed -> Search Vector DB -> Get Chunks ->

Send to GPT -> Answer.

- **Video:** RAG Architecture Explained

Thursday: "Add Your Data" Feature

Practice: Use the built-in "Chat with your data" feature in the Studio for quick prototyping.

Friday: Optimization

Practice: Experiment with "Chunk Size". Does 500 characters work better than 2000?

✅ WEEK 11: Azure AI Foundry (Prompt Flow)

Goal: Enterprise Orchestration.

Monday: Prompt Flow Visuals

Practice: Open AI Foundry. Clone a "Standard Flow". Run it.

- **References:** Prompt Flow Tutorial

Tuesday: Custom Python Tools

Practice: Create a Python node in the flow that fetches the current time or calls a weather API.

Wednesday: Evaluation

Practice: Run a "Groundedness" eval to measure if the AI answer matches the context.

- **References:** Evaluation Metrics for GenAI

Thursday: Deploying Flows

Practice: Deploy the flow to a Managed Endpoint (just like Week 6).

Friday: Agents Intro

Practice: Use the Assistants API to create an agent with a Code Interpreter.

- **References:** AI Assistants Concept

✅ WEEK 12: CAPSTONE PROJECT

Goal: Prove you are ready for the job.

Schedule: * **Mon-Tue:** Design. Draw the architecture. Select the tools (e.g., Form Recognizer + GPT-4 + SQL).

- **Wed-Fri:** Build. Code the frontend (Streamlit) and backend (FastAPI/Azure).
- **Sat:** Test. Break it. Fix it.

- **Final Monday:** Demo.

Choose Your Path: 1. **HR Resume Matcher:** Upload PDF resumes -> Extract Skills -> Match with Job Description -> Rank Candidates.

2. **Support Ticket Auto-Solver:** Read Email -> Classify Intent -> Search Knowledge Base -> Draft Reply.

3. **Financial Analyst Agent:** Read CSV data -> Write Python code to analyze trends -> Summarize in English.